

Monotone Memory-Depth Optimization

A First-Passage Theorem for Nested Weyl Certificates

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Abstract

Finite-memory support certificates trade the cost of applying old normalized Gram snapshots against a geometric bound for the omitted tail. We prove that this trade is ordered more rigidly than a generic sequence of matrix approximations: if each newly admitted increment is paid for by the exact decrease in the tail budget, then the Weyl lower bound for any relative singular mode is monotone in memory depth. The first depth crossing a target is therefore the cost-minimal certificate, and searching through the full available history is complete for this certificate family. A single-path five-scale audit tests every depth on 360 threshold-labelled updates. All 322 genuinely supported records are certified, with no tail, dominance, monotonicity, minimality, or completeness failures. The largest required depth is six; the aggregate snapshot-action cost is reduced by more than 72 percent relative to full history. These are finite operator results and do not establish an all-level physical support law.

1 Nested projected crosses

Let $V : \mathbb{R}^r \rightarrow \mathbb{R}^n$ be an isometry and $P_\perp = I - VV^*$. Fix a target cross

$$C = P_\perp G V,$$

and write a sequence of recent-window approximants as

$$C_{d+1} = C_d + A_d, \quad d = 1, \dots, N-1.$$

The discarded part has an operator budget $\delta_d \geq 0$. We require the increment condition

$$\|A_d\| \leq \delta_d - \delta_{d+1}, \quad \delta_{d+1} \leq \delta_d. \quad (1)$$

For the k th relative singular mode, define the nested Weyl lower

$$L_d^{(k)} = \frac{(s_k(C_d) - \delta_d)_+}{s_1(C_d) + \delta_d}. \quad (2)$$

Whenever $\|C - C_d\| \leq \delta_d$, the singular-value perturbation inequalities imply

$$L_d^{(k)} \leq \frac{s_k(C)}{s_1(C)}$$

[Horn and Johnson, 1991].

The point of RH-116 is that the values in (2) are not merely valid one at a time. Under (1), they are ordered.

Theorem 1 (Nested Weyl monotonicity). *Suppose (1) holds. Then, for every available singular mode,*

$$L_{d+1}^{(k)} \geq L_d^{(k)}.$$

Proof. Weyl's inequalities give

$$s_k(C_{d+1}) \geq s_k(C_d) - \|A_d\|, \quad s_1(C_{d+1}) \leq s_1(C_d) + \|A_d\|.$$

Consequently,

$$(s_k(C_{d+1}) - \delta_{d+1})_+ \geq (s_k(C_d) - \delta_d)_+,$$

while

$$s_1(C_{d+1}) + \delta_{d+1} \leq s_1(C_d) + \delta_d.$$

The numerator is nondecreasing and the positive denominator is nonincreasing, proving the claim. The zero-matrix case follows by the zero convention in (2). $\square$

This theorem does not assert that the raw singular values of C_d are monotone. They need not be: projected positive Gram increments can cancel one another directionally. The monotone object is the lower endpoint after the newly resolved increment is removed from the uncertainty budget.

2 Geometric memory and first passage

For normalized Gram snapshots $S_j \succeq 0$ with $\text{tr } S_j = 1$ and $0 \leq \eta < 1$, consider

$$G = \sum_{a=0}^{N-1} \eta^a S_{t-a}, \quad C_d = P_\perp \left(\sum_{a=0}^{d-1} \eta^a S_{t-a} \right) V.$$

Since $\|S_j\| \leq 1$, the newly included cross obeys $\|A_d\| \leq \eta^d$. The finite geometric tail

$$\delta_d = \sum_{a=d}^{N-1} \eta^a = \eta^d \frac{1 - \eta^{N-d}}{1 - \eta} \tag{3}$$

satisfies $\delta_d - \delta_{d+1} = \eta^d$. Theorem 1 therefore applies exactly.

Corollary 2 (Cost-minimal first passage). *Let $\tau > 0$ and let*

$$d_*(\tau) = \min\{d : L_d^{(k)} \geq \tau\}.$$

If a depth- d action costs rd snapshot–packet products, then the ascending search may stop at its first pass, and $d_(\tau)$ has minimum cost among all passing depths.*

Proof. The cost is strictly increasing in d . By Theorem 1, every later depth also passes, so the first passing depth is precisely the least costly one. $\square$

Corollary 3 (Full-history completeness). *If $C_N = C$ and $\delta_N = 0$, then an ascending search through $d = N$ finds a certificate if and only if $s_k(C)/s_1(C) \geq \tau$.*

Proof. Every passing lower is dominated by the target ratio. Conversely, $L_N^{(k)} = s_k(C)/s_1(C)$, so a supported target passes at the final depth. $\square$

Completeness here is deliberately narrow: it concerns this nested Weyl family. Failure at full history correctly reports lack of target support, but failure of an early depth does not rule out a different kind of certificate.

3 Single-path five-scale audit

RH-115 found that separately rounded Gram assemblies can disagree after division by a weak fourth mode. The present audit removes that interface. For each scale, side, threshold, and time, every normalized snapshot is formed once. Full and truncated Gramians are then accumulated by the same newest-to-oldest binary64 routine. The packet refresh uses that same full Gramian. Thus depth comparison never transports a lower bound between two independently assembled operator representatives. The archived audit also checks the actual discarded cross norm against (3); outward rounding conventions follow standard interval logic [Moore, 1966].

The data contain five scales, two sides, three thresholds, and 120 physical updates, giving 360 threshold-labelled records. All available depths from one through the complete history are enumerated even though Corollary 2 permits early stopping. This supplies an independent audit of monotonicity, minimum depth, and full-history completeness.

threshold	supported	certified	largest depth	mean cost/full cost
10^{-8}	115	115	6	0.374
10^{-6}	109	109	5	0.351
10^{-4}	98	98	4	0.314

Table 1: Adaptive first-passage results out of 120 updates per threshold.

Across all thresholds there are 322 supported records and 322 adaptive certificates. There are zero failures of tail enclosure, lower-bound dominance, monotonicity, minimum-depth identification, and full-history completeness. Even before applying a tolerance, no floating-point decrease of $L_d^{(4)}$ is observed. All 78 fine-chain updates pass at each threshold.

The first-passage histogram is

d	1	2	3	4	5	6
10^{-8}	18	33	31	25	7	1
10^{-6}	18	32	29	25	5	0
10^{-4}	18	31	26	23	0	0.

At $\eta = 1/512$, the infinite tail budget drops by a factor of 512 per added snapshot, from 1.95695×10^{-3} at depth one to 5.56198×10^{-17} at depth six. The mean recordwise cost is 0.348 of full history. Aggregating the rank-weighted snapshot actions gives a cost reduction exceeding 72 percent.

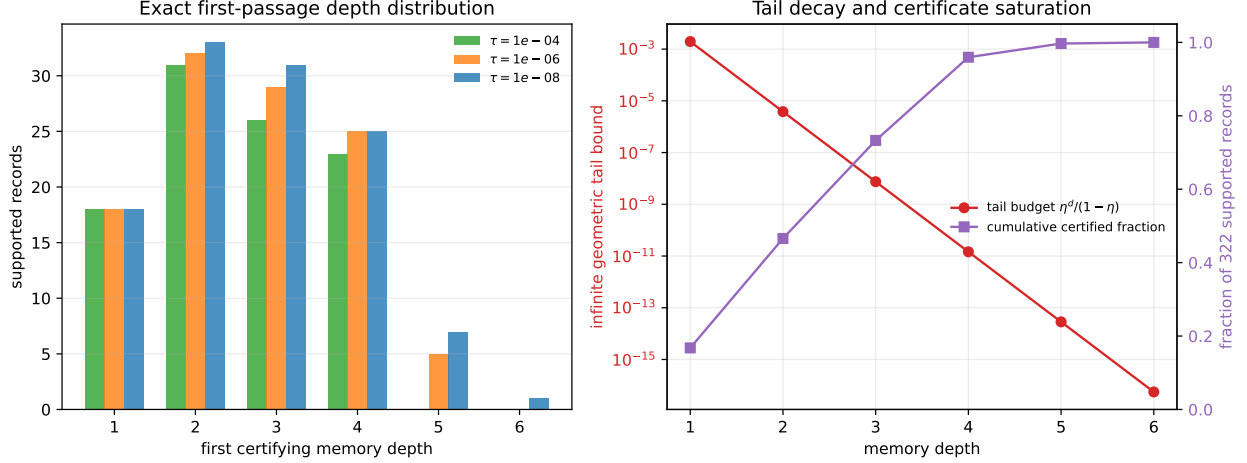


Figure 1: Left: exact first-passage depths. Right: geometric tail decay and cumulative saturation of the 322 supported records.

4 Consequence and boundary

The theoretical gain is the monotone first-passage principle: memory depth is now an exactly optimizable certificate parameter, not a heuristic tuning choice. The numerical gain is that one consistent assembly recovers every supported archived record, including the two weak 10^{-8} records not closed by the fixed-depth RH-110 audit.

The largest observed depth six is not promoted to an all-level theorem. Doing so would require physical lower margins and scale laws that persist beyond the five anchors. RH-117 therefore audits those scale trends and proves what finite anchor data cannot imply. Nothing here constructs a Hilbert–Polya operator, identifies zeta zeros, closes uniform Stage A, or proves the Riemann Hypothesis.

References

- Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.
- Ramon E. Moore. *Interval Analysis*. Prentice-Hall, 1966.